

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Monday, 24 August 2026, 9:04 PM**State** Finished**Completed on** Monday, 24 August 2026, 9:05 PM**Time taken** 35 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int find_sum_pair(int *arr, int low, int high, int x, int *e1, int *e2) {
5      if (low >= high) {
6          return 0;
7      }
8
9      int current_sum = arr[low] + arr[high];
10
11     if (current_sum == x) {
12         *e1 = arr[low];
13         *e2 = arr[high];
14         return 1;
15     }
16
17     if (current_sum > x) {
18         return find_sum_pair(arr, low, high - 1, x, e1, e2);
19     } else {
20         return find_sum_pair(arr, low + 1, high, x, e1, e2);
21     }
22 }
23
24 int main() {
25     int n;
26     if (scanf("%d", &n) != 1) return 0;
27
28     int *arr = (int *)malloc(n * sizeof(int));
29     if (arr == NULL) return 0;
30
31     for (int i = 0; i < n; i++) {
32         if (scanf("%d", &arr[i]) != 1) {
33             free(arr);
34             return 0;
35         }
36     }
37
38     int x;
39     if (scanf("%d", &x) != 1) {
40         free(arr);
41         return 0;
42     }
43
44     int e1 = 0, e2 = 0;
45     if (find_sum_pair(arr, 0, n - 1, x, &e1, &e2)) {
46         printf("%d\n%d\n", e1, e2);
47     } else {
48         printf("No\n");
49     }
50
51     free(arr);
52     return 0;

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course